

Clark Review: Editorial Holding Area

This companion preserves the original working list. Some items are now integrated, some require source checks, and others remain alternatives. The current destination and status map is in `editorial/STRUCTURE_AUDIT.md`; the preserved pre-reorganization draft is in `editorial/archive_2026-09-24_before_restructure/`. These notes are not assertions that every proposed criticism has been established. In particular, old instructions to perform work may have been superseded by the completed analyses documented in the audit.

Working Notes: Points to Revisit

- **Parent and grandparent averages in Table A1.** Revisit the erroneously labelled average-parent and average-grandparent entries in the book's technical appendix. Distinguish a correlation with an average, a regression slope on that average, and an average of individual-relative correlations. Averaging changes reliability and the normalization, so the common individual attenuation factor cannot simply be carried over. Check the practical relevance separately: these average rows do not appear among the Table 4.5 moments specified for estimation.
- **Intuition for nonidentification of measurement error under genetic assortment.** Explain why independent true environmental variation and independent recording noise both dilute the observed genetic signal without affecting mate selection on genetic value. The correlations reveal only the product θh^2 : less reliable measurement of a more heritable trait is observationally equivalent to more reliable measurement of a less heritable trait. Even the observed spouse correlation does not separate the two in this model.
- **The fundamental measurement-error exclusion.** An imperfect measure of a latent trait is not automatically that trait plus classical measurement error. The gap between observed education and an asserted latent social ability may reflect family resources, access to schooling, preferences, discrimination, and sex-specific opportunities. These influences may covary with the person's own genetic and environmental components, relatives' components, and relatives' measurement residuals. Our common- θ correction sets these covariances to zero. If they are nonzero, additional terms enter the observed kinship covariances and a uniform attenuation factor need not apply. Develop this objection: an admissible estimate of θ does not validate the exclusions or establish that an error-free heritability has been recovered.

- **Identification with generation-specific measurement error.** Revisit identification when attenuation differs across generations rather than being governed by one common θ . How can the difference between full-sibling and parent–child correlations then distinguish the lineal factor $(1 + r)/(1 + m)$ from different attenuation factors? The same question arises for the single-grandparent moment relative to the aunt/uncle moment. Specify whether reliability varies by historical generation or by the number of generations separating a pair, derive the resulting pair-level attenuation factors, and establish what additional restrictions or moments would separate measurement error from the effects of r versus m .
- **Measurement correction added to A1–A3, but parameter recovery unclear.** The book inserts θ into the moment equations, so the A3 intercept identifies $\exp(a) = \theta h^2$, not latent heritability alone. Yet the surrounding text still describes estimated heritability and cites the PNAS SI Table S2. Check whether the plotted/reported quantity is θh^2 or a separately recovered h^2 . Under phenotypic assortment, recovery from A3 is possible using $m = 2b - 1$, $r = (1 + m) \exp(c) - 1$, $h^2 = m/r$, and $\theta = \exp(a)/h^2$, subject to admissibility; retaining the same regression is not itself an identification error. Distinguish this passage from Chapter 5’s use of Table A2’s spouse and affine-relative moments for Denmark and Sweden. Under the stated common-attenuation formulas, the sibling-in-law/sibling ratio identifies r , while the spouse correlation identifies θr , providing an additional route to separate reliability and heritability. Audit that estimation and Table 5.4 separately rather than assuming the PNAS procedure applies.
- **Reconcile Figure A1, PNAS Figure 1, and PNAS SI Table S2.** The book’s Figure A1 (p. 298) is the counterpart of PNAS Figure 1, but several plotted coordinates differ. Approximate graphical readings of the heritability axis put higher education 1860–1919 near 0.58 in PNAS and 0.33 in the book, and occupational status 1860–1919 near 0.66 and 0.53, respectively. PNAS Figure 1 also disagrees with its own SI Table S2: occupational status 1780–1859 is plotted near 0.94, whereas the table reports 0.722. These are graphical readings, not recovered numerical estimates. Establish the source data, estimation version, and parameter interpretation for each display; do not infer from the changed points alone that measurement error or the structural constraint was newly handled correctly. For an independent extension, the book’s Table 5.1 (p. 67) reports observed education correlations for spouses and several affine relatives in Denmark and Sweden. Keep these distinct from the English education cohorts and from the latent correlations inferred in Table 5.2.
- **An apparent impasse between assortment interpretations.** With phenotypic assortment and free measurement attenuation, the lineal contrast and $m = h^2 r$ allow separate recovery of h^2 and θ . However, our unweighted A3 fits to Table 4.5 yield inadmissible structural recoveries for four of eight outcomes: $r > 1$ for company directorship, and $h^2 > 1$ for occupational status 1780–1859, higher education 1780–1859, and literacy. Alternatively, if the intended assumption is genotypic assortment, the moment formulas must change: the separate lineal adjustment disappears ($c = 0$),

and these kinship correlations identify only m and θh^2 . Intergenerational observations can improve estimation and test the common decay pattern, but do not add the identifying contrast needed to separate h^2 from θ . Thus one cannot rescue the interpretation merely by switching assortment assumptions while retaining separately identified heritability. Is this an impasse for the interpretation of the reported estimates? Distinguish that concern from a formal rejection: bounded phenotypic estimation remains possible, the results depend on weighting and sampling uncertainty, and externally justified information about reliability or heritability could resolve the genotypic model's nonidentification.

- **Claims about regression to the mean.** Otto et al. explicitly discuss linear regression toward the mean in cultural models, including the possibility of reproducing the genetic model's rate. Clark is therefore wrong to present linear regression toward the mean as a distinct prediction of genetic transmission. Distinguish the shape of the regression relationship from its numerical rate, and distinguish a fixed prediction from a fitted parameter: in the genetic benchmark at issue the coefficient is 0.5, whereas in the cultural model it is a free parameter, which can also take that value. This is a difference in predictive restriction, not a demonstration that cultural transmission cannot generate the same pattern. Supply the explicit passages from Otto et al. and Clark, and specify the assumptions behind the genetic 0.5 benchmark rather than treating it as universal.
- **Key points to hammer home.** There is genuine value in Clark's analyses, and the review should acknowledge that explicitly. The central concern is not that the analyses are without merit, but whether the conclusions he draws follow logically from them. Preserve this distinction when assessing both contributions and limitations. Often Clark's conclusions do not follow from the evidence he presents. This does not establish that the conclusions are wrong; it means they have not been established nearly as credibly as he suggests. Develop this distinction consistently throughout the review. Working analogy to consider: his apparent resistance to alternative explanations recalls, in the opposite direction, the stubbornness of Lysenkoite social scientists who instinctively reject evidence allowing any role for genetics. The target is asymmetric standards of evidence, not an equivalence of substantive views. He also sometimes exaggerates how strongly his tests discriminate between genetic and cultural models; symmetry and regression toward the mean are key examples to document. Examine his claims that school reforms yield no benefits, distinguishing evidence of no benefit from failure to establish a benefit. Track shifts between policy criteria: does a reform help anyone or improve average outcomes, does it help the poor, does it reduce inequality, and does it increase or reduce mobility? These are distinct questions, and a failure on one criterion does not establish failure on the others. Collect specific passages showing where the criterion changes within an argument.
- **Selective discussion of assumptions in twin and adoption studies.** Clark's discussion focuses on one assumption whose violation can bias estimated h^2 downward, but these designs depend on other assumptions as well. Identify the assumption he

emphasizes and examine the others, including violations that could bias estimates upward or have an ambiguous effect. Distinguish twin from adoption designs and assess the direction of each bias under explicit conditions. A possible source of downward bias alone does not establish that the estimates are lower bounds on heritability.

- **Lower resemblance when raised apart and claims about adoptees.** Add a discussion of the evidence for lower resemblance among relatives raised apart than among those raised together, and what this comparison implies for Clark's interpretation of environmental influences. His conclusions about adoptees appear much stronger than the evidence warrants. Identify the precise claims and supporting studies, report the relevant comparisons and uncertainty, and distinguish what those studies establish from broader claims about the absence of effects of rearing environments.
- **Within-MZ-twin estimates are not automatically causal.** Acknowledge Clark's valid point that comparisons within monozygotic twin pairs do not by themselves identify causal effects. Controlling for shared genes and family background does not eliminate nonshared confounding or measurement error. Also note that he does not cite the prior literature documenting these limitations, either within economics or outside it. Supply the relevant references and check his discussion and bibliography when developing this point: the methodological concern is important, but it is not new.
- **Implications for education spending and social outcomes.** Discuss the claims: "First implication... is that modern societies spend too much on education." "What drives much of this expenditure is a belief that boosting years of education by less advantaged students increases social mobility rates." "Second implication... social outcomes will change only when the genetic endowment of the society changes." This implies that maximum genetics explains 60%.
- **The covariance exclusions matter beyond genetic nurture.** Revisit Clark's rejection of genetic nurture on p. 86, but do not frame the objection as depending on this particular causal mechanism. Any relevant covariance that the fitted model incorrectly sets to zero can bias the recovered genetic, assortment and attenuation parameters. This includes environmental covariance between relatives and cross-person genetic-environmental covariance beyond the terms already implied by the stated assortment model. Establishing or rejecting a specific "nature of nurture" pathway does not establish the required zero-covariance restrictions. Write out the omitted covariance terms and show how they enter the fitted moments; do not presume that every violation biases every parameter, or that the direction is always the same. The issue is the validity of the identifying exclusions, not the label attached to the source of resemblance.
- **Binary phenotypes and the scale of the model.** Examine whether the linear phenotype, kinship-correlation and common measurement-error formulas apply to binary outcomes such as higher education, literacy and company directorship. Distinguish

correlations and heritability on the observed binary scale from those of a latent liability. Investigate whether prevalence, thresholds, and differences across cohorts or sexes alter the predicted moments, attenuation and identification. In a threshold/liability model, the observed Pearson correlation is generally a nonlinear function of the latent correlation and the two marginal prevalences. Check explicitly whether one common attenuation factor θ can represent that transformation across kinship types; generally it need not. Compare a properly specified threshold model with the direct fit to binary Pearson correlations, and check whether Clark supplies such a link anywhere in the book or PNAS materials. Work out the implications before treating a binary outcome as a continuous phenotype with classical measurement error. The potentially serious problem is that thresholding changes the pattern of correlations across kinship types, not merely their level. Even if Fisher's model fits the latent correlations perfectly, its geometric decay restrictions generally fail for the observed binary Pearson correlations; fitting the original formulas can then distort inferred heritability and assortative mating. Demonstrate this using jointly normal liabilities whose population correlations satisfy the model exactly. Threshold them and compare fits to population tetrachoric correlations (the binary case of polychoric correlations), which recover the latent correlations under these assumptions, and to Pearson correlations. At 50% prevalence in both groups, the exact mapping is $r_{\text{binary}} = (2/\pi) \arcsin(r_{\text{latent}})$: latent correlations of 0.8 and 0.4 become approximately 0.590 and 0.262, so even their ratio changes. Use several kinship types to show why neither a common attenuation factor nor refitting a single persistence rate generally restores the exact fit. The problem is especially consequential for the θ extension: classical measurement error motivates a common multiplicative attenuation, whereas thresholding generally implies an effective $\theta_k = r_{\text{binary},k}/r_{\text{latent},k}$ that varies across kinship types and marginal prevalences. In the example above, these ratios are approximately 0.738 and 0.655. On the log scale, $\log \theta_k$ is therefore not a common intercept shift: it can alter the fitted kinship-decay slope and lineal-collateral contrast used to recover the structural parameters. Allowing one free θ does not generally repair this misspecification; allowing a separate unrestricted θ_k for every kinship type would instead remove the cross-kinship restrictions that make the model informative. A threshold model should derive these transformations from prevalences and latent correlations. This population example isolates measurement-scale misspecification from sampling error; quantify its practical importance at the prevalences and correlations relevant to Clark's data.

- **Fit competing cultural and genetic transmission models.** Assess the absence of a serious comparison with joint cultural-genetic transmission or a purely cultural model. A good fit of the restricted genetic model, even if established, does not identify that mechanism. Specify and compare the alternatives under transparent assumptions and the same data and objective, distinguishing predictive restrictions from extra flexibility. For the regression-to-the-mean point above, Otto et al. (1994, pp. 8–9, eq. 8) write

$$B_o = \beta_I(B_m + B_f) + \delta S_C,$$

where S_C is cultural transmission error, not schooling. The equation resembles ge-

netic transmission, but β_I is not fixed at one half. They explicitly describe regression towards the mean under cultural transmission; this pattern alone cannot select the genetic explanation.

- **Disconnect between behavioural genetic estimates and policy conclusions.** Develop the distinction between variance-component estimates and the causal effects of feasible interventions. Fixed genes do not imply fixed effects, and a statistical direct genetic effect need not operate through an unmodifiable physiological pathway. Use the eyeglasses example and relate the point to Goldberger (1978) and Jencks (1980). Give explicit credit for the provocative questions, valuable dataset and interesting patterns while explaining why the model-selection and policy conclusions go beyond what those results establish. Engage with the empirical evidence rather than personal labels.
- **Assortment on “something deeper” requires a model.** Address Clark’s argument that people assort on something deeper than the observed phenotype. Specify what that latent characteristic is, how it relates to observed outcomes and their genetic and environmental components, and how it governs partner selection. Derive the spouse covariances and kinship moment conditions under that matching rule, including the measurement assumptions needed to connect latent and observed outcomes. An appeal to a deeper trait is not sufficient to retain the original moment formulas or parameter interpretations. If this is simply phenotypic assortment on a latent trait measured with classical error, state and justify those restrictions explicitly; if it is a different assortment mechanism, establish which moments and identification results change.
- **Selective treatment of bias and lower within-family GWAS estimates.** Examine whether Clark acknowledges and adequately addresses the much lower estimates reported in within-family GWAS. His emphasis on biases that could depress other estimates should be assessed alongside evidence that controlling for family background reduces population GWAS effect estimates or polygenic-score associations. Identify the relevant studies, report the comparable within-family and population estimates with their uncertainty, and check his discussion before asserting an omission. Distinguish SNP effects, score associations or prediction, and SNP-based heritability from total narrow-sense heritability; these are not interchangeable quantities. Assess explanations involving indirect genetic effects, population structure, assortment and measurement error under explicit assumptions, rather than treating the lower estimates as either automatically decisive or automatically dismissible.
- **Use within-family PGIs in spousal analyses.** Clarify that analyses intended to interpret spousal PGI resemblance as assortment on direct genetic effects should use PGIs constructed with within-family GWAS effect estimates. Population-GWAS weights can also reflect population structure and indirect genetic associations, so spousal correlations in those scores need not isolate the intended genetic component. Compare the results using population and within-family weights, allowing for differences in precision and predictive accuracy. Within-family weighting is not

a complete solution: establish what the score measures, which biases remain, and how its spousal correlation relates to the model's latent additive-genetic correlation m .

- **Misspecification need not produce poor fit.** Make explicit that a misspecified model can fit the observed moments well: fitted parameters may absorb omitted covariance terms, and different mechanisms may generate similar patterns of resemblance. Consequently, good fit does not validate the identifying restrictions or the genetic interpretation of the estimates. Keep this logical point separate from the empirical concern that fit may itself be poor once the model's restrictions and admissible parameter bounds are imposed. Assess both, but do not make the identification criticism depend on establishing poor fit or a formal statistical rejection.
- **Residual stratification despite ancestry restrictions and quality control.** Document the substantial GWAS evidence that population stratification can bias estimates even after standard quality control and restriction to an apparently homogeneous ancestry group. Such restrictions do not establish the absence of fine-scale population structure or its association with social and environmental differences. Assemble concrete examples, specifying the phenotype, ancestry definition, controls and evidence used to diagnose residual bias. Explain the implications for interpreting population-GWAS effects and spousal PGI correlations as purely genetic quantities. Distinguish stratification from indirect genetic effects and assortative mating; a within-family versus population difference alone need not identify which mechanism accounts for it.
- **The Collado–Clark debate and the identification disputes of the 1970s.** Explore the parallels with the debates involving Jencks, the Birmingham group, and Cloninger and colleagues. A recurring problem is identification of distinct transmission mechanisms from a small set of kinship moments: similar fits can conceal very different causal models and parameter interpretations. Check the historical positions and the precise points of disagreement before drawing the comparison. The irony is how much progress has occurred in data, molecular measurement and research designs, while this basic identification problem can reappear when inference rests on a limited set of resemblance patterns. Explain which newer evidence can actually distinguish the competing mechanisms, rather than treating technical progress itself as a solution to identification.
- **Goldberger on adding environments to Fisher's scheme.** Use Goldberger's pointed observation, preserving its specific target: "The Birmingham models gain empirical validity by tacking on environmental correlations to Fisher's scheme" (Goldberger, 1978, p. 36). Discuss the accompanying coherence problem: adding a shared-environment term to sibling resemblance requires an account of the process generating it and its implications for other kinships. Goldberger asks whether shared sibling environmental variance should reappear for uncles and cousins, what equates parent–child and sibling environmental shares in another

specification, and why parental transmission of both environments and genes does not generate gene–environment covariance. Follow his Appendix to spell out the restrictions for a coherent extension. Do not turn this into a claim that every shared sibling influence must persist across generations; the implications depend on how the environment is generated and transmitted.

- **Modern evidence on dominance does not establish zero shared environment.** Section VI now develops the distinction between dominance and gene–environment covariance. Extend its outcome-specific evidence where appropriate. Identify the relevant findings, state exactly which claim about dominance they support, and delimit the outcomes and designs to which they apply. This does not vindicate the separate claim that shared-environment variance c^2 is zero for all outcomes in adulthood. Evaluate that claim against outcome-specific evidence and uncertainty, keeping dominance and shared-environment components conceptually distinct. Support for one modelling assumption cannot stand in for evidence for the other.
- **Read Alex Young’s commentary on Clark.** Read Young’s initial reaction and extended July 2023 commentary on Clark’s PNAS paper, including his discussion of alternative transmission models, assortative mating and identification. Compare the arguments with our review before deciding whether to cite or address them. The posts are dated 26 June and 4 July 2023; original links are retained in the saved source notes. Copies and retrieval notes are saved in the project reference library.
- **Add analogy to equilibrium restriction from economics.** Explain why $m = h^2r$ is a consequence of the maintained phenotypic-assortment model, not an optional additional assumption. Select an economic analogy that distinguishes primitive assumptions from their implied restrictions, and distinguishes estimating an unrestricted model to test the restriction from interpreting its fit as the fit of the maintained model.

Preferred option: the consumption Euler equation. Consider a consumer with logarithmic utility who can transfer resources between two periods at gross return R , under certainty and with an interior optimum. Utility maximisation implies $c_2/c_1 = \beta R$, where β is the discount factor. Consumption growth, patience and the return cannot be specified independently while preserving that model. The Euler equation is a consequence of the optimisation problem, not an optional additional assumption. The assortment restriction plays an analogous role: freeing m , r and h^2 requires changing the maintained model, rather than simply declining to impose one of its consequences.

Preferred option: homogeneity of consumer demand. In the standard consumer model, doubling all prices and nominal income leaves the budget set unchanged. With unchanged preferences and a unique optimum, it therefore leaves demand unchanged. Homogeneity of demand is a consequence of the model, not an additional assumption to be accepted or discarded independently. A fitted demand equation that responds to this common rescaling may be useful descriptively, but it cannot represent

the maintained choice model. Likewise, the phenotypic-assortment model does not permit independent choices of m , r and h^2 : its assumptions require $m = h^2 r$.

- **Cite the Georgia land lottery paper.** Cite Bleakley and Ferrie (2016), “Shocking Behavior: Random Wealth in Antebellum Georgia and Human Capital Across Generations,” on the 1832 Cherokee Land Lottery, when discussing exogenous wealth shocks and intergenerational outcomes.
- **Regression towards the mean does not identify the transmission mechanism.** Focus the criticism of Chapter 14 on the inference from an approximately linear conditional mean to a claim about its causal mechanism. The figures do show substantial regression towards the mean; do not base the critique on denying that descriptive finding. A nearly flat child-on-parent relationship, as for longevity, means especially strong regression. The dashed lines in the empirical panels are fitted lines, not the identity line.

Show explicitly that a purely cultural model can generate the same linear, symmetric pattern: if $C_o = \gamma(C_f + C_m)/2 + U$, with an independent mean-zero social innovation U , then the conditional mean given the parental average is linear. Under suitable Gaussian assumptions and innovation variance, this model can reproduce the one-generation joint distribution of parental and offspring phenotypes in our additive Fisher specialisation. This is a counterexample to identification from these plots, not a claim that every extended-pedigree model is equivalent. Assess Clark’s conclusion that the pattern confirms the likely dominance of direct genetic transmission (p. 216) against a specified cultural alternative, rather than equating cultural transmission generally with the particular poverty/privilege-trap pattern sketched on p. 211.

Supporting qualifications: genetic additivity alone does not guarantee an exactly linear conditional mean; linearity, contraction and symmetry about a common population mean require distinct assumptions; the high reported R^2 values for grouped means are not individual-level explanatory power or formal tests against nonlinearities; and symmetry of log deviations is not equality of ordinary percentage changes. Keep these qualifications subordinate to the identification criticism. The source-checked quotations, graphical slope estimates, identity-line replots and mathematical counterexamples are saved in the Chapter 14 audit dated 23 September 2026 in the project notes.

- **Chapter 18: equilibrium variance and a numerical correction.** With fixed segregation variance S , the chapter’s equation (6) implies $V(m) = 2S/(1 - m)$, hence $V(0.6)/V(0) = 2.5$, not the fivefold increase stated on p. 274; the fivefold result is correct for the correlation of 0.8 used in Clark and Cummins (2022, pp. 3–4). The discrepancy is consistent with an uncorrected carryover from the working paper, although the revision history has not been verified. For the final review, emphasize that this is a standard additive-genetic equilibrium result under a constant-segregation-variance approximation, whereas interpreting latent social ability as

genetic, relating genetic to phenotypic assortment ($m = h^2r$ under phenotypic homogamy, with potentially changing h^2), and attributing the Industrial Revolution to the enlarged upper tail require separate justification. The increase in equilibrium genetic variance under positive assortative mating is a classical result of quantitative genetics, originating in Fisher's framework and developed more explicitly in subsequent theoretical work. Clark's contribution here lies in its proposed historical application, whose empirical premises and causal interpretation require separate evaluation. Trace the result through Fisher, subsequent formal treatments such as Nagylaki's, and textbook accounts, distinguishing Fisher's explicit results from later clarifications rather than attributing its origin to a modern exposition.

- **Cousin marriage does not establish weak genetic assortment.** Examine Chapter 18's inference from cousin marriage to a low equilibrium spousal genetic correlation (pp. 278–281). Random choice of a cousin with respect to ability or status is an assumption, not an implication of cousin marriage; selection within the permitted kinship pool could change the result. More fundamentally, establish whether the cousin correlation $[(1 + m)/2]^3$ remains valid under recurrent consanguineous mating before accepting the fixed-point calculation $m = [(1 + m)/2]^3$: repeated related mating creates additional shared ancestors and paths of genetic relationship that require an explicit pedigree or population-genetic treatment. Treat this as a derivation to check, not as a completed demonstration of the corrected equilibrium. Distinguish genetic relatedness through shared ancestry from correlation in additive genetic values for a particular trait. The calculation alone does not establish weaker genetic assortment in the historical societies concerned, nor their comparative supply of exceptional innovators.

References

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- Goldberger, A. S. (1978). Models and methods in the IQ debate: Part I. revised. SSRI Workshop Series Paper 7801, Social Systems Research Institute, University of Wisconsin–Madison.
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- Otto, S. P., Christiansen, F. B., and Feldman, M. W. (1994). Genetic and cultural inheritance of continuous traits. Manuscript dated 28 July 1994; distinct from the 1995 working-paper citation.